AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM276
Software Development Processes and Methodologies

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 20 /Apr./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part I – Multiple Choice Questions — 3 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. Which general issue that affect software describes the statement "Systems are required to operate across networks that include different types of computers":
   a. Heterogeneity b. Business and Social Change c. Security and Trust d. Scale

2. Which software is assembled from existing components:
   a. Waterfall b. Incremental development c. Integration and configuration d. V-model

3. Which extreme programming principle states that requirements should be recorded and used to determine when to be implemented based on priority.
   a. Incremental planning b. Small releases c. Pair programming d. On-site customer

Part II – Short Answer Questions — 12 Marks

*Answer **all** the following short answer questions. Each question is worth 4 Marks.*

1. A small family-owned business decided to create a new software to help them manage their work. Which software process model would be suitable to use? Justify your answer.

2. Discuss the below Extreme Programing practices
   a. Incremental planning
   b. Small releases

3. Given the below User Requirement definition, provide 2 System Requirements Specification.
   The system shall enforce a strong password creation policy when customers are creating new accounts.

Part III – Problem Solving Questions — 15 Marks

*Answer **all** the following problem-solving questions. Each question is worth 7.5 Marks.*

1. Develop an activity diagram for a Mechanic checking errors and warnings stored on the On-Board Computer (OBD) System of a car. Make sure to cover the following requirements:
   a. Use Swimlanes to represent 2 roles, the Mechanic and the OBD System.
   b. The Mechanic will start the task to check for warnings, safety errors, and non-safety errors, which can be done simultaneously. In case safety errors were found, the mechanic has to report them, otherwise the errors should be reset on the OBD. At the end, a report should be printed.
   c. Make sure to use at least 8 activities.
   d. Make sure to use Synchronization bars and Decision and Merge nodes.

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2. Consider the battery monitoring system of a mobile phone. Generate the state machine for the class BatteryStatus with 3 states (Full, Acceptable, Low).
   a. Draw the state machine diagram and add the 3 states
   b. For each state add at least 1 self-transition
   c. For each state add at least 1 transition to another state
   d. The events to be used must be only "Used" and "Charged"
   e. Add Guards when needed.

End of questions

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